

Hearth Companion

A home companion for an older person living alone — it sees, talks, runs her errands within limits her family set, and asks her before it ever alarms anyone.

The idea: bounded agency. AI agents are about to start acting — and spending — for people. Hearth acts freely within limits, stops and asks a relative above them, and brings a human in only when it matters. Elder care is the case; the pattern is general.

WHAT IT DOES — THREE FLOWS

FLOW 1 • MONEY

Groceries over budget

"Order my groceries for the week."

Hearth builds her usual basket (€84). It's over the €60 weekly limit, so it **stops and asks**: Claire gets the basket and the excess on Telegram with **Approve / Reject**. One tap approves exactly that purchase, once, for 15 minutes → Stripe charge → Hearth tells Marie, with the delivery time. A second tap charges nothing.

FLOW 2 • SIGHT

It can see her

"Can you see me? Do I have my glasses on?"

The camera's latest frame and a one-line scene description travel with every question, so Hearth answers from what it actually sees. It only talks about the room when asked — never narrates on its own, never identifies people.

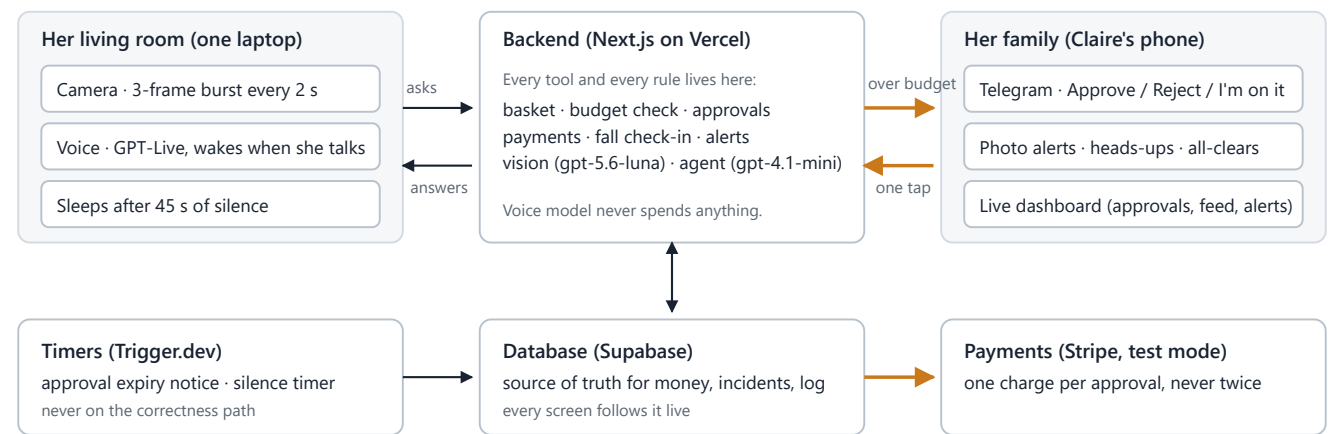
FLOW 3 • FALL

Asks first, then family

"Marie? I noticed you're on the floor. Are you okay?"

Two consecutive "on the floor" readings (~4 s) → a heads-up to Claire → Hearth **speaks first**. "I'm fine" closes it. Silence or "I'm not okay... yes" → Telegram alert with a **photo of the room** and an "I'm on it" button. A false alarm costs one spoken question.

HOW IT WORKS



WHAT HAPPENS WHEN SHE FALLS — THE TIMELINE

t = 0 s camera sees her on the floor (sequence of frames, not a still – a fall, not sitting down)
t ≈ 4 s second confirmation → incident opened · Claire gets "⚠️ Possible fall – checking with Marie now"
t ≈ 5 s Hearth speaks first: "Marie? I noticed you're on the floor. Are you okay?"
t ≈ 5-25 s listens
 "I'm fine, I dropped my glasses" → resolved · Claire gets "✅ False alarm"
 silence, or "I'm not okay / help / yes" → alert with room photo + [I'm on it]
later Claire taps I'm on it → Hearth: "Claire has seen it and is on her way."

THE RULES — WRITTEN IN CODE, NOT LEFT TO THE AI

- **The camera never alarms directly.** Two confirmations to ask; escalate only on silence, a request for help, or an explicit "yes".
- **The vulnerable person is never the sole approver of a large spend.** Anything over her weekly limit goes to a relative.
- **One approval, one charge.** An approval is consumed in a single atomic database statement; a second tap, an expired approval, or a changed basket charges nothing. Payments are idempotent.
- **Budget read from the database every time** — a limit changed on the dashboard applies immediately.
- **Emergency services are never called automatically.** A human decides.
- **Sight is answered, not narrated.** It never claims it can't see, and never identifies anyone — it describes what's visible.
- **Privacy.** Frames are analysed for an event and not stored; only the frame attached to an alert leaves the house — to the family.
- **Designed for a real impaired user:** no blinking lights, nothing to press, repeated questions answered as kindly the fortieth time as the first.

THE SCREENS

SCREEN	FOR	WHAT IT SHOWS
/room	Marie	Dark, calm, huge text. No buttons. Wakes when she talks: Resting · Listening · Thinking · Speaking · Checking on you.
/monitor	Caregiver / demo	Live camera, current state, 5-step workflow (Surveillance → Chute détectée → Attente → Appel → Alerte), event journal, conversation log, demo controls, load a recorded clip.
/family	Claire	Pending approvals with countdown and Approve/Reject, live activity feed, alerts, and an assistant ("What did Mom order this week?").

BUILT WITH

PIECE	WHAT	WHY
OpenAI	GPT-Live voice · gpt-5.6-luna vision · gpt-4.1-mini tools	The voice model holds the conversation and <i>delegates</i> work to our backend — tools and money stay under our rules.
Supabase	Postgres + Realtime	Source of truth; the database is the event bus every screen follows.
Vercel	Next.js app	Public HTTPS for the Telegram webhook; deploys on every push.
Telegram	Family channel	Approvals, heads-ups, photo alerts, one-tap buttons.
Stripe (test)	Payments	Real charge mechanics against a simulated shop.
Trigger.dev · CopilotKit	Timers · dashboard assistant	Expiry notice and silence timer; "what did Mom order?"

LINKS

Live	https://hearth-companion-cyan.vercel.app	Code	https://github.com/Loudiyii/hearth-companion
Screens	/room · /monitor · /family	Built at	AI Tinkerers "Agents, Everywhere" — Paris, 12 Sept 2026

Hearth Companion · Act within limits · Ask above them · Bring a human in only when it matters. Stripe is test mode against a simulated shop; emergency calls are never automatic.